

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 2– Seminar / Poster Presentation (Emerging LLMs & Industry Applications)**

|                         |                                                                                                                                                        |                      |                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------|
| <b>Course Code:</b>     | CSA65                                                                                                                                                  | <b>Course Title:</b> | Generative AI and Large Scale Models                                                     |
| <b>CO Assessed:</b>     | CO5- BL-5 – Articulation                                                                                                                               | <b>Assessment:</b>   | Assessment Tool 2- Seminar / Poster Presentation (Emerging LLMs & Industry Applications) |
| <b>Weightage:</b>       | 10% of CIA                                                                                                                                             | <b>Total Marks:</b>  | 1 * 100= 100                                                                             |
| <b>CO4 Statement:</b>   | <i>Deploy Generative AI applications while evaluating ethical, security, governance, and responsible AI considerations in industrial environments.</i> |                      |                                                                                          |
| <b>Student Name:</b>    | S.Sirivally                                                                                                                                            | <b>Reg. No.:</b>     | 192472371                                                                                |
| <b>Section / Batch:</b> | 2024                                                                                                                                                   | <b>Date:</b>         | 29-08-26                                                                                 |

### Instructions to Students

- Poster Presentation – Instructions

#### 1. Poster Format

- Poster Size: A1 or A2
- Orientation: Portrait
- Poster should be clear, attractive, and visually engaging.
- Use a large, readable title and appropriate font sizes.
- Use diagrams, flowcharts, images, graphs, and icons instead of lengthy text.

#### 2. Poster Content

- Each poster should contain the following sections:
- Title of the Topic
- Participant Name(s) & Department
- Introduction / Background
- Problem Statement
- Overview of the LLM Technology
- Architecture / Workflow / Working Principle
- Industry Applications
- Real-World Use Case
- Advantages

- Challenges / Limitations
- Future Scope
- Conclusion
- References
- Duration 30 mins

### **Code of Conduct**

*I S.Sirivally / 192472371 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_Sirivally .S \_\_

---

## **SECTION A : POSTER PRESENTATION**

---

### **◆ Advanced & Research-Oriented Topics**

---

#### **1. Federated Learning and Privacy-Preserving LLMs**

---

# FEDERATED LEARNING AND PRIVACY-PRESERVING LLMs

Secure Collaborative Intelligence Without Centralizing Sensitive Data

S.Sirivally – 192472371  
B.E- CSE (AI)

## INTRODUCTION

- Large Language Models (LLMs) require huge amounts of data for training and fine-tuning.
- Much of this data can contain sensitive or private information.
- Centralizing such data creates privacy and security risks.
- Federated Learning enables collaborative model training without sharing raw data.
- Each organization keeps its data locally and shares only model updates.



**Key Idea:** Data stays with the owner. Only model updates are exchanged and aggregated.

## PROBLEM STATEMENT

- Traditional LLM training requires collecting and processing huge datasets at a centralized location, which can expose sensitive personal, medical, financial, or organizational information.
- Many organizations cannot directly share their datasets because of privacy regulations, data ownership, security risks, and confidentiality requirements.
- Centralized training also creates a single point of failure, while data from different organizations may have different formats, distributions, and privacy requirements.
- Therefore, there is a need for a collaborative approach where organizations can improve a common LLM without transferring their raw private data to a central server.

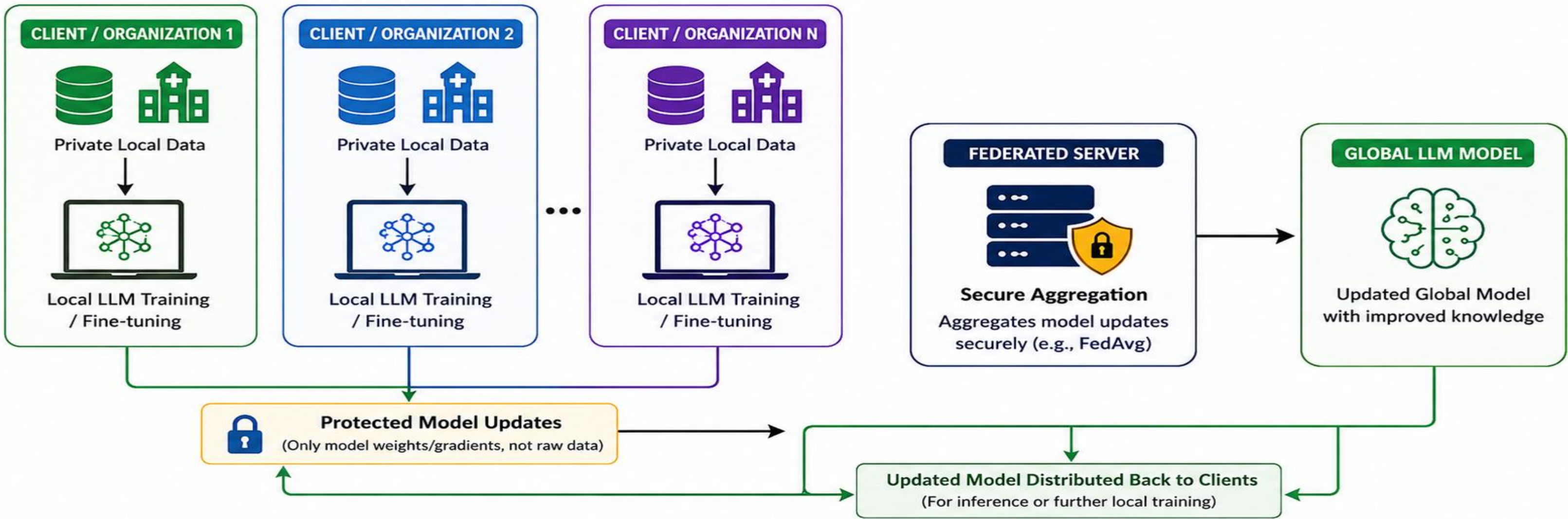
## OVERVIEW OF LLM TECHNOLOGY

- LLMs are deep-learning models trained on large text datasets to understand and generate human-like language.
- They perform tasks such as text generation, question answering, summarization, translation, code generation, and reasoning.
- LLMs can be adapted to specific domains using fine-tuning and PEFT methods such as LoRA.
- In Federated Learning, LLMs are trained locally while only model updates are shared, protecting sensitive data.

### REAL-WORLD CASE:

- HEALTHCARE** — Hospitals can collaboratively improve a medical LLM while keeping patient records within each institution.
- FINANCE** — Banks can collaborate on shared AI capabilities while retaining sensitive customer and transaction data locally.

## ARCHITECTURE



## BENEFITS AND LIMITATIONS

### ADVANTAGES / BENEFITS

- Data Privacy:** Raw sensitive data remains with the organization, reducing the risk of data exposure.
- Collaborative Learning:** Multiple organizations can jointly improve an LLM without directly sharing their private datasets.

### LIMITATIONS / DISADVANTAGES

- High Communication Cost:** Frequent exchange of model updates between clients and the server can require significant network resources.
- Computational Complexity:** Training large LLMs locally requires substantial computing power and can be slower than centralized training.

## CONCLUSION & REFERENCES

### CONCLUSION

- Federated Learning enables multiple organizations to collaboratively train or fine-tune LLMs without sharing their raw private data.
- Combining Federated Learning with privacy techniques helps achieve secure, collaborative, and privacy-preserving AI development.

### FUTURE SCOPE

- Efficient Federated Fine-Tuning:** Develop faster and more resource-efficient methods for training large LLMs across distributed devices.
- Stronger Privacy & Scalability:** Improve privacy protection and make federated LLMs practical for large-scale healthcare, finance, enterprise, and edge applications.

### REFERENCES

- McMahan et al., “Communication-Efficient Learning of Deep Networks from Decentralized Data,” AISTATS, 2017.
- Kairouz et al., “Advances and Open Problems in Federated Learning,” Foundations and Trends in Machine Learning, 2021.
- Li et al., “Federated Learning: Challenges, Methods, and Future Directions,” IEEE Signal Processing Magazine, 2020.

## RUBRICS FOR CALCULATION

| Criteria                 | Wt. | Level 4 (Exemplary)                                                | Level 3 (Proficient)                 | Level 2 (Developing)                | Level 1 (Beginning)                    |
|--------------------------|-----|--------------------------------------------------------------------|--------------------------------------|-------------------------------------|----------------------------------------|
| Currency of Content      | 30% | Demonstrates current, accurate knowledge of emerging trends/models | Mostly accurate and current content  | Some outdated or inaccurate content | Largely inaccurate or outdated content |
| Industry Relevance       | 25% | Clearly connects trends to relevant engineering industry use-cases | Reasonable industry connections made | Weak industry connections           | No industry relevance shown            |
| Research Quality         | 20% | Well-researched with credible, recent sources                      | Adequately researched                | Limited or dated sources used       | Unresearched content                   |
| Visual & Delivery Impact | 15% | Visually strong poster/slides; confident delivery                  | Clear poster/slides; decent delivery | Basic visuals; weak delivery        | Poor visuals and delivery              |
| Citations & Takeaways    | 10% | Proper citations and clear takeaways for the audience              | Mostly cited; clear takeaways        | Minimal citation; unclear takeaways | No citations; no clear message         |

### Marks Evaluation:

| Question No.        | Currency of Content (30%) | Industry Relevance (25%) | Research Quality (20%) | Visual & Delivery Impact (15%) | Citations & Takeaways (10%) | Total Marks (100) |
|---------------------|---------------------------|--------------------------|------------------------|--------------------------------|-----------------------------|-------------------|
| Q1                  |                           |                          |                        |                                |                             |                   |
| Overall Total (100) |                           |                          |                        |                                |                             |                   |